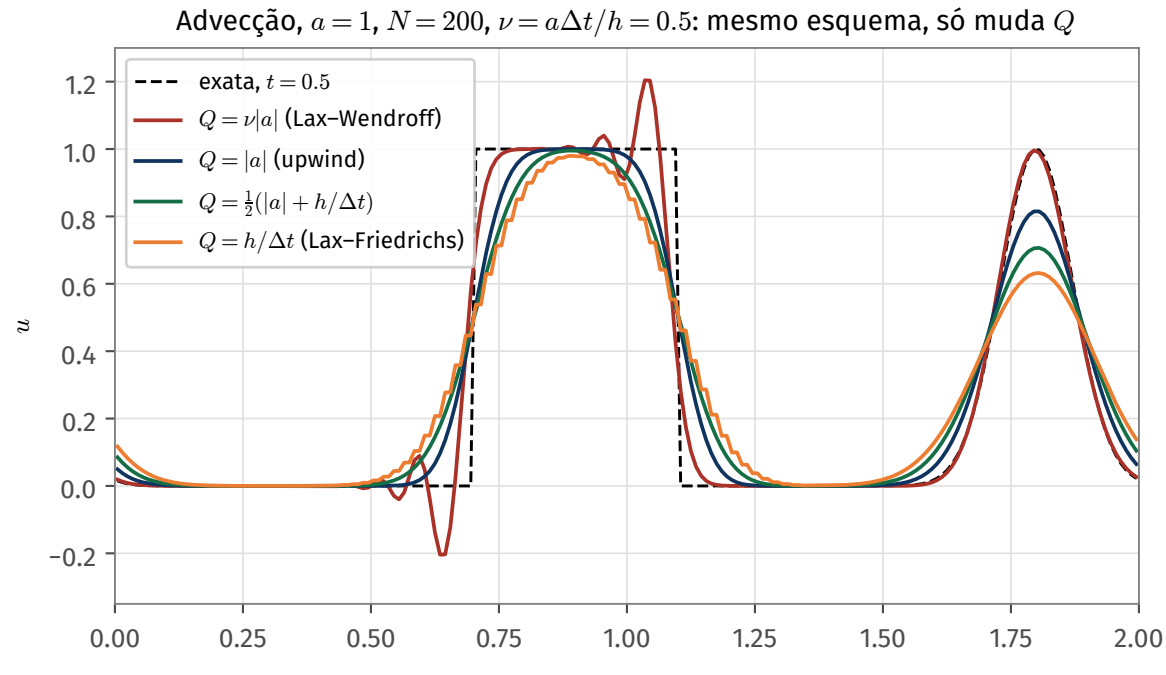
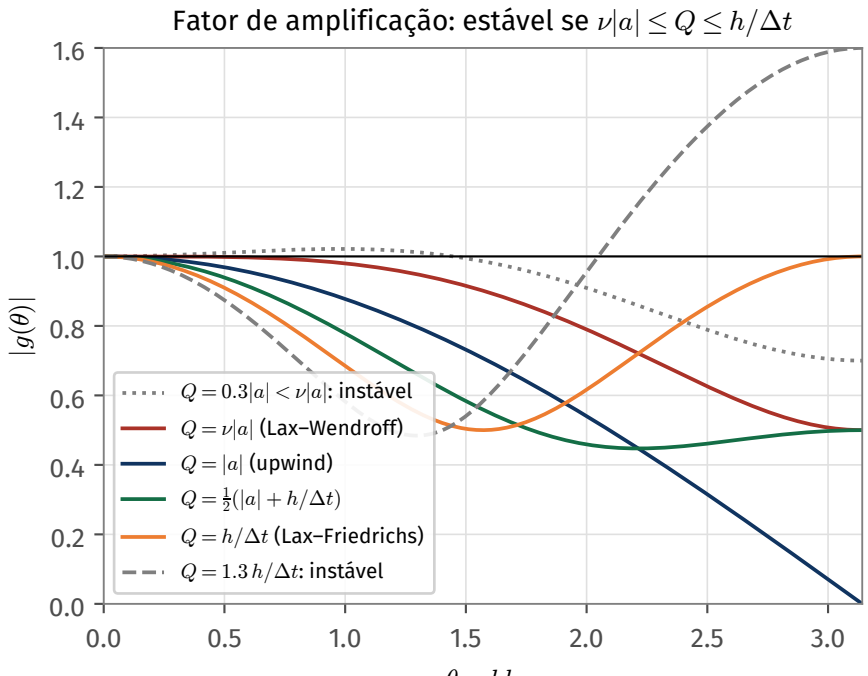

Advecção, $a = 1$, $N = 200$, $\nu = a\Delta t/h = 0.5$: mesmo esquema, só muda Q
exata, $t = 0.5$
$Q = \nu|a|$ (Lax–Wendroff)
$Q = |a|$ (upwind)
$Q = \frac{1}{2}(|a| + h/\Delta t)$
$Q = h/\Delta t$ (Lax–Friedrichs)
u
x


Fator de amplificação: estável se $\nu|a| \le Q \le h/\Delta t$
$Q = 0.3|a| < \nu|a|$: instável
$Q = \nu|a|$ (Lax–Wendroff)
$Q = |a|$ (upwind)
$Q = \frac{1}{2}(|a| + h/\Delta t)$
$Q = h/\Delta t$ (Lax–Friedrichs)
$Q = 1.3\,h/\Delta t$: instável
$|g(\theta)|$
$\theta = kh$